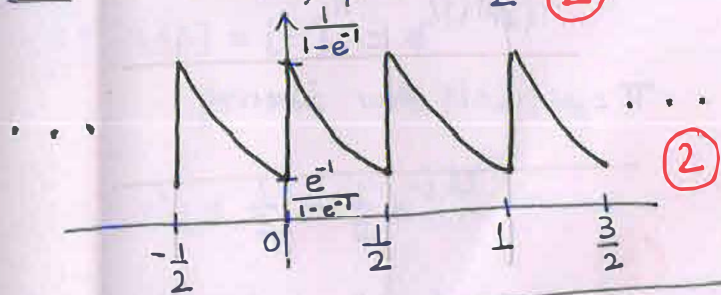


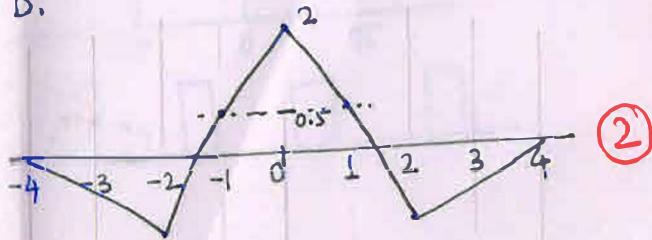
$$y(t) = \frac{e^{-2t}}{1-e^{-1}} \quad 0 < t < \frac{1}{2}$$

Q1 a. Periodic; period $\frac{1}{2}$ (2)

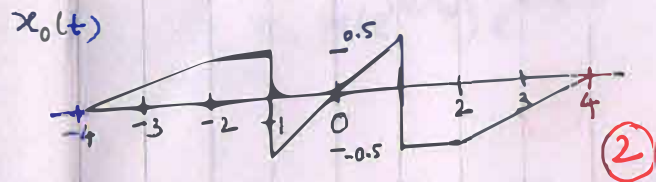


$$\text{Power} = \frac{1}{2} \frac{e+1}{e-1} \quad (2)$$

Q2 b.



Sketch $x(t)$ (1)



Q1 c.

$$g(t) \rightarrow g(2-t) \quad (4)$$

$$y(t) = \int_{t-4}^t e^{-3(t-\tau)} x(\tau) d\tau \Rightarrow \int_{-\infty}^t e^{-3(t-\tau)} x(\tau) d\tau - \int_{-\infty}^{t-4} e^{-3(t-\tau)} x(\tau) d\tau$$

Q2. Impulse response of

$$\text{System 1: } h_1(t) = e^{-3t} [u(t) - u(t-4)] \quad (2)$$

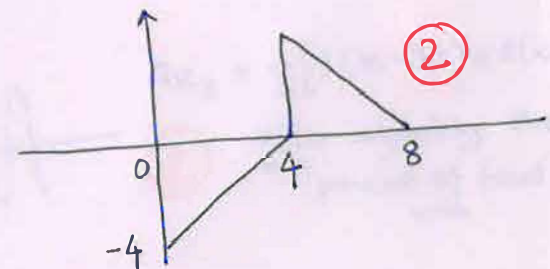
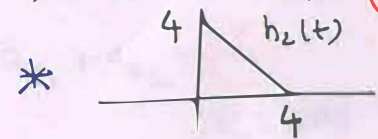
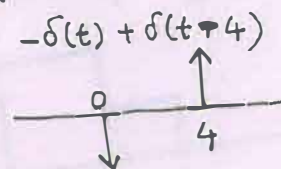
a. Causal, stable, memory (1)

$$b. y(t) = x(t) * h_1(t)$$

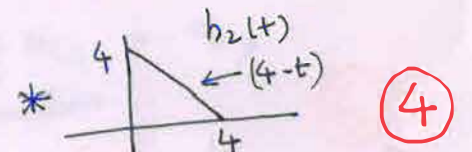
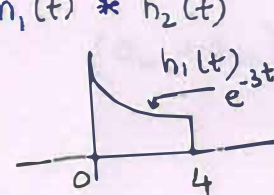
$$= 5e^{-2} \left\{ \left[-e^{-3(t-1)} + e^{-2(t-1)} \right] u(t-1) \right\} - e^{-12} \left\{ e^{-3(t-5)} + e^{-2(t-5)} \right\} u(t-5) \quad (4)$$

c. Impulse response of

$$\text{System 2: } h_2(t) = (4-t)(u(t) - u(t-4)) \quad (2)$$



$$d. h_1(t) * h_2(t)$$



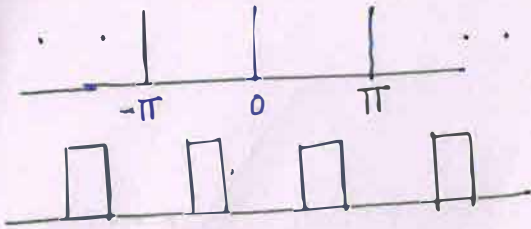
Q3

(a) $x_1[n] = [-1]^n = e^{j(2\pi/2)n}$

periodic with $N=2$; $\omega_0 = \pi$

$$a_k = \frac{1}{2} - \frac{1}{2} e^{-j\frac{2\pi}{2}k}$$

$$a_0 = 0; a_1 = 1$$

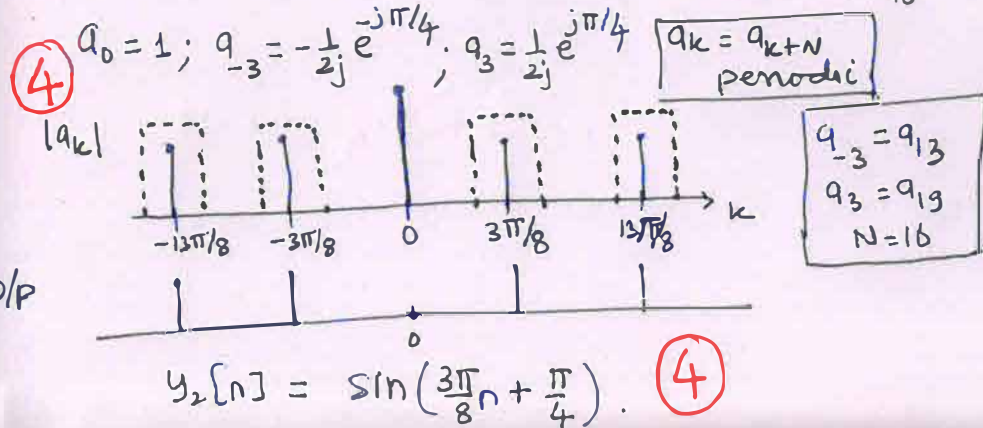


zero.

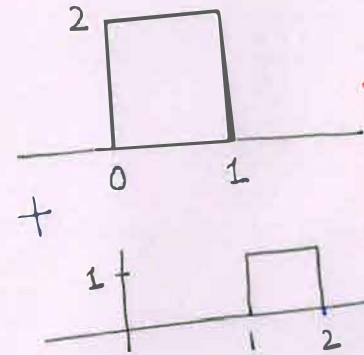
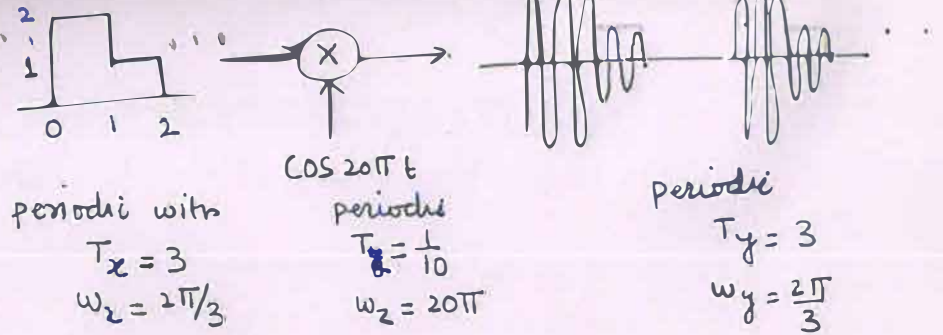
$$y_1[n] = \sum_{k \in \langle N \rangle} a_k H(e^{jk\omega_0}) e^{jk\omega_0 n}$$

$$= 0$$

(b) $x_2[n] = 1 + \sin\left(\frac{3\pi}{8}n + \frac{\pi}{4}\right)$; $\omega = \frac{3\pi}{8} = 2\pi \cdot \frac{3}{16}$
 $N=16$ $\omega_0 = \frac{2\pi}{16}$



Q4



$$a_{k_1} = \frac{2}{j\pi k} + \delta(k)$$

$k: \text{odd}$

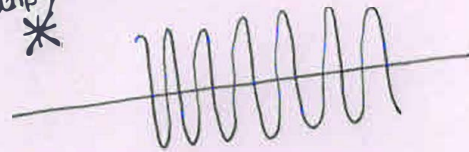
$$a_{k_2} = \frac{e^{-jk\omega_0(1)} \cdot 1}{j\pi k} + \frac{1}{2} \delta(k)$$

$k: \text{odd}$

$$a_{k_1} + a_{k_2} = (4)$$

$k: \text{odd}$

multiply *



$$a_{k_3} = \frac{1}{2} [\delta(k-30) + \delta(k+30)]$$

(2) after adjusting the period of final signal with

$$(a_{k_1} + a_{k_2}) * a_{k_3} = a_k y$$

convolution

$k: \text{odd}$

(6) final answer